Detect Driving Behavior and Attitude

(HARIS حريص)

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Summary

This innovative project aims to revolutionize road safety by developing an AI-powered application named High Alert Recognition Intelligent System (HARIS) that detects and analyzes driver and driving behaviors using camera in-car and road cameras. The core objective is to create a safer driving environment by reducing accidents caused by reckless driving and driver distractions. Through innovative computer vision techniques, the application intelligently analyzes images in real-time to identify a wide range of critical driver behaviors, such as aggressive lane changes, and different unsafe driver behavior operated while driving. The Detect Driver Distraction Model utilizes the YOLOv5 object detection model to detect and classify multiple distraction classes, achieving an impressive accuracy of 0.95% Precision. The app accurately classifies and interprets these behaviors, allowing for timely notifications and alerts to be sent to the driver. The Driving Behavior Detection Model employs YOLOv8 and ByteTrack algorithms to detect various car behaviors and road lanes, achieving a substantial MAP of 0.78% and 0.70% on training and validation respectively at a threshold of 0.5. Additionally, the system utilizes YOLOv5 and a CRNN-based OCR model to detect and recognize car license plates, attaining a line accuracy of 0.91%. These capabilities allow for the identification of potentially dangerous vehicles. The results demonstrate the effectiveness of HARIS in accurately detecting driver distractions, monitoring driving behavior, and identifying risky situations on the road, thereby contributing to improved road safety, and ensuring a safer journey for everyone on the road.

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1. Introduction

Driver behavior is the description of intentional and unintentional characteristics and actions a driver performs while operating a motor vehicle. It can be influenced by several factors like age, experience, attitude, fatigue, etc. These internal and external factors can change even the same driver’s ability to assess risk and make driving decisions from situation to situation. and driving a vehicle is an indispensable part of everyday life for many people. However, sometimes this everyday life does not go as expected, as many accidents happen on public roads, and most of these accidents are due to driver behavior. Whereas 90% of traffic accidents are caused due to driving errors. Human errors due to inattention, distraction, mental workload, and poor observation are the most frequent (41%) [1], Here the importance of artificial intelligence becomes clear as researchers and developers aim to create safer and more efficient driving experiences.

This project focuses on the development of the High Alert Recognition Intelligent System (HARIS), an AI-based solution that utilizes computer vision to analyze and interpret both driver behavior and driving behavior. Computer vision is a field of artificial intelligence that focuses on extracting information from visual data, such as images or videos.

The primary objective of the HARIS system is to detect and analyze driver distractions and driving behaviors using in-car cameras and road cameras. By gaining a comprehensive understanding of these behaviors, the system can effectively identify potential dangers and provide timely warnings and alerts to the driver. The aim is to enhance driver safety by mitigating risks based on the analysis of their behavior behind the wheel.

The HARIS system consists of two main models designed to perform specific computer vision tasks. The first model, known as the HARIS Driver Distraction Detection model, focuses on detecting and classifying various driver behaviors. By analyzing images or videos captured by the in-car camera, the system can accurately identify behaviors such as safe driving, texting, talking to passengers, drinking, etc. and whether the driver is wearing a seatbelt or not. by Leveraging YOLOv5 that updated versions have enhanced accuracy and speed, making them even more ideal for real-time distraction detection. The You Only Look Once (YOLO) method is one of the most promising computer vision techniques for object detection. YOLO is a convolutional neural network-based object detection and classification method that can recognize and categorize objects in real-time video streams. Using it in real-time assessments of the driver's actions and behaviors can be made.

The second model, the HARIS Driving Behavior Detection model, is specifically designed for detecting and analyzing various car behaviors using visual data from road cameras. This model is designed to identify indicators of unsafe driving, including lane changes then recognition of car license plates. By leveraging YOLOv8 with ByteTracking for detect car and road lane and assign ID to each car and lane, then use YOLOv8 to detect car plate of violator vehicle, then use Convolutional Recurrent Neural Network (CRNN) with Connectionist Temporal Classification (CTC) loss to recognize car plate character and numbers, this model aims to contribute to a safer driving experience and promote responsible driving practices.

One of the objectives of this project is to develop and integrate the HARIS models into a mobile application, providing real-time notifications to drivers and assisting them in making safer and more efficient driving decisions. The other objective is to automatically notify the government or relevant authorities about the dangerous behavior, by generating reports to facilitate timely intervention and ensure road safety.

In the following sections, we will discuss the technical details of the project, its potential benefits and limitations, and its impact on the future of driver behavior analysis as follows. In Section 2 methodology is presented including datasets and selected models used and details on process followed along conducting this project, while Section 3 explains the Results and analysis . Afterwards, conclusions and recommendations are developed in Section 4. Finally, in Section 5 References. Through this project, we aim to advance the field of driver behavior analysis and pave the way for safer roads in the future.

2. Methodology

2.1. Related Work

2.1.1. HARIS Driver Distraction Detection Related work.

- Seatbelt vs No Seatbelt classification project by Sachin_raj on Kaggle [2].

The researcher has collected data from youtuber who make content related to cars.

1. seatbelt-contain yolov5 model and labeled data.
2. seatbelt2-contains data for training of a seatbelt vs no seatbelt classifier.
3. seatbelt3-contains both trained yolov5, classifier and test video.

- Driver Distraction Behavior Detection Method Based on Deep Learning by researchers Peng Mao, Kunlun Zhang, Da Liang [3]

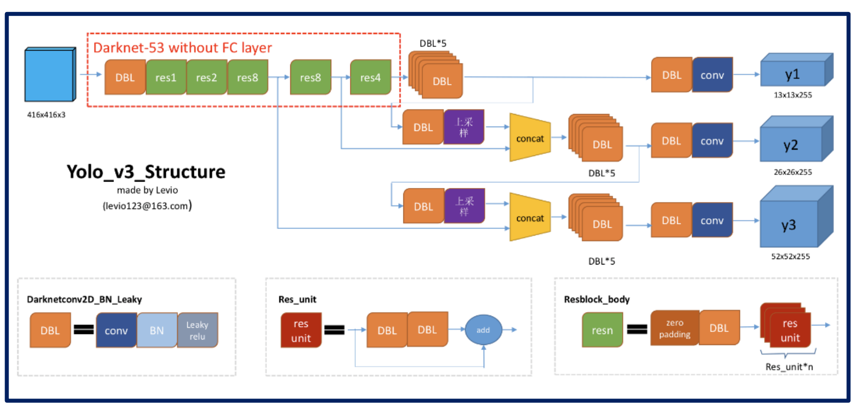


Figure 1:YOLOv3 structure

The researcher made several modifications to the YOLOv3 model shown in Figure 1. Instead of using SoftMax for classification, which only allows for single-class statistics, they introduced an independent logistic regression classifier. This classifier determines whether the object within the target box belongs to the current label, enabling a simple two-class classification.

Additionally, the researcher incorporated the concept of Feature Pyramid Network (FPN) into YOLOv3. They added sampling and convolution modules to the base network, generating multi-scale feature maps for detection. This approach enhances the model's ability to capture semantic information from images at various levels. Each feature map predicts only three bounding boxes, resulting in a total of nine bounding boxes for the detection task. The initial widths and heights of these bounding boxes are obtained through clustering.

Finally, the trained YOLOv3 detection algorithm was utilized to detect distracting behaviors such as smoking and calling, providing a practical application of the modified model.

2.1.2 HARIS Driving Behavior Detection and car plate recognition Related work.

Automatic Number-Plate Recognition Image Processing Group Project Summary 2022/23 Fall Semester [4]

Extracting characters' sequences or texts from images is an early research problem as many researchers tried to address it in many research works. There are several types of algorithms used to solve this research dilemma. This part of the project discusses different methods (algorithms) used to address the characters' sequence recognition problem.

In (Automatic Number-Plate Recognition Image Processing Group Project Summary 2022/23 Fall Semester) paper they start with transfer learning to train a YOLOv7 based on the original implementation. They detect a single object in each image: license plates. Trained model for 100 epochs. They achieved around 90% precision on both training, data, and validation data, for OCR they initially followed the OCR architecture proposed by Shi et al. (2017) but had to switch to the paddleOCR system by Du et al. (2020) due to unsatisfactory results. The Shi et al. (2017) method is a CRNN-based approach that combines DCNN and recurrent neural networks (RNN) for end-to-end sequence recognition. It utilizes LSTM layers to capture long-range dependencies in image-based sequences and achieves improved transcription accuracy compared to commercial OCR engines.

After cutting the license plate, they apply the following preprocessing methods with OpenCV: Normalization, Noise removal, Erodation, Blur clarification and Adaptive Threshold. In Figure 2 exhibits various natural-looking weather effects, including a Snow filter and a Sun filter, along with their immediate results. The original image, after being cut out from the object detection step, is presented alongside the preprocessed version, which serves as input for the subsequent Optical Character Recognition.

Figure 2: Several types of natural-looking weather effects (Snow and Sun filter)

2.2. Datasets

2.2.1 HARIS Driver Distraction Detection Datasets.

- Distracted Driving Computer Vision Project [5]

Distracted Driving Dataset is an open-source dataset published by the Ipylot project. It is available on the Roboflow Universe platform is Published in 2022, it contains driver images captured within a car while the driver is performing various tasks, such as talking on the phone, texting, reaching behind, Drinking, and more.

Dataset contains about 7000 images in training set, 1000 in validation and 1000 in test sets. There are 12 classes labeled from 0 through 11shows in Table 1 and their distribution in Figure3. That shown in the Table1:

Table 1:Class Labels and their Names.

Label	Class Name
c0	Safe Driving
c1	Texting
c2	Talking on the phone
c3	Operating the Radio
c4	Drinking
c5	Reaching Behind
c6	Hair and Makeup
c7	Talking to Passenger
d0	Eyes Closed
d1	Yawning
d2	Nodding Off
d3	Eyes Open

The distribution of each class for different districted behavior appears in Figure 3:

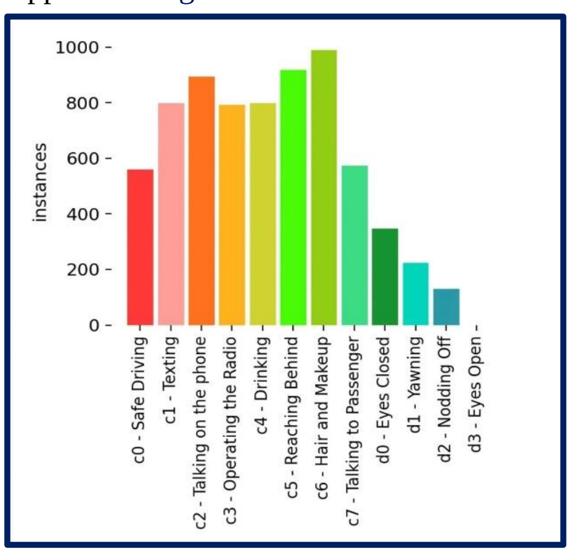


Figure 3 :Class distribution of various driving distractions

- “Seatbelt_data_1” [6]

The seatbelt detection dataset used consists of two classes: "seatbelt" and "no seatbelt." In Table 2. The dataset maintains a balanced distribution between the two classes, ensuring equal representation for seatbelt and no seatbelt instances. The images are labeled with appropriate class identifiers, indicating the presence or absence of seatbelts.

Table 2 : Class Labels and their Names.

Label	Class Name
0	No seat belt worn
1	Seat belt worn

This dataset was obtained from Kaggle, a popular platform for sharing and accessing datasets. It encompasses a diverse range of images that were meticulously annotated in YOLO dataset format. The dataset was collected by the publisher from relevant YouTube videos, which was then transformed into a comprehensive dataset.

2.2.2 HARIS Driving Behavior Detection and car plate recognition Datasets.

- License Plate Text Recognition Dataset [7]

The License Plate Text Recognition Dataset available on Kaggle is a collection of license plate images specifically created for the task of license plate text recognition. The dataset was generated using an object detection model that initially identified license plates in a larger dataset. These license plates were then cropped based on the bounding box coordinates provided by the object detection model. Dataset contains 20,000 images in jpg format with a corresponding label file in csv format. We used this dataset for training and validation of our license plate recognition model. In Figure 4 shows sample from this dataset.

Figure 4:Sample from License Plate Text Recognition Dataset.

- Car License Plate Detection [8]

Our car license plate detection project utilized a dataset of 433 images in (.png) file Figure 5, specifically focused on car license plates. The dataset annotated the output car plates detection by cropping the bounding box (each with bounding box annotations in XML format.) and writing the plates labels in csv file. We used this dataset for testing our license plate recognition model.

Figure 5:Example of Car License Plate Detection dataset

- Road Lane Instance Segmentation Image Dataset [9]

The Road Lane Instance Segmentation Dataset is specifically curated for lane instance segmentation tasks, enabling the detection of road lines using lane demarcation lines. With a collection of 5398 images, the dataset is split into 4983 training samples, 311 validation samples, and 104 test samples Figure 6. It serves as a valuable resource for developing computer vision models that can accurately identify and track road lanes from camera inputs in various road scenarios.

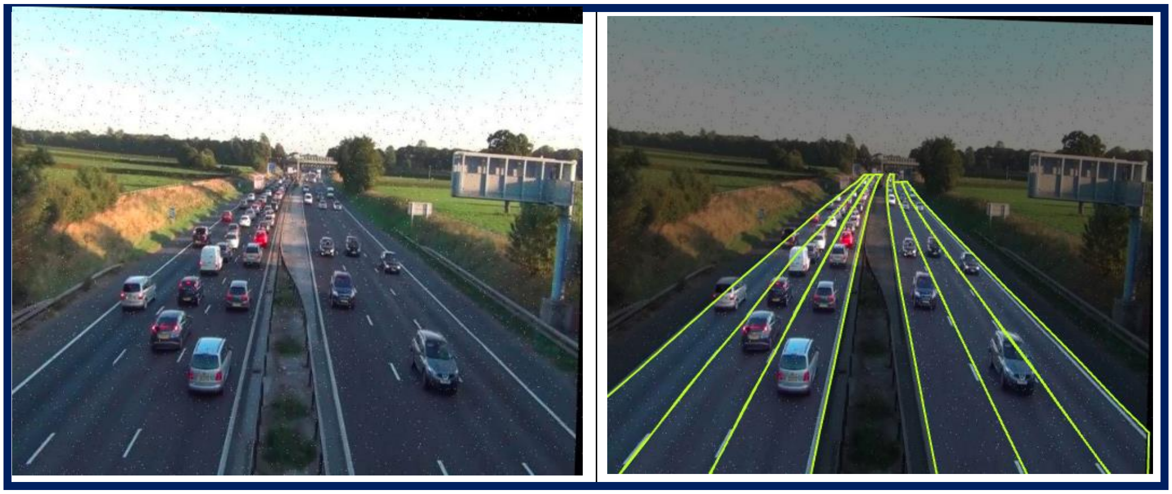


Figure 6:Sample from Road Lane Instance Segmentation Image Dataset.

- Common Objects in Context (coco) dataset [10]

The COCO dataset is one of the most popular large-scale labeled image datasets available for public use. It represents a handful of objects we encounter daily and contains image annotations in 80 categories shown in Figure 7, with over 1.5 million object instances. It's used for multiple CV tasks: Object detection and instance segmentation, Image captioning, Key points detection, Panoptic segmentation, Dense pose, and Stuff image segmentation.

The below image represents a complete list of 80 classes that COCO has to offer. Specifically, we selected car categories and track categories to detect. According to our project, these categories are referred to as "vehicles".

person	fire hydrant	elephant	skis	wine glass	broccoli	dining table	toaster
bicycle	stop sign	bear	snowboard	cup	carrot	toilet	sink
car	parking meter	zebra	sports ball	fork	hot dog	tv	refrigerator
motorcycle	bench	giraffe	kite	knife	pizza	laptop	book
airplane	bird	backpack	baseball bat	spoon	donut	mouse	clock
bus	cat	umbrella	baseball glove	bowl	cake	remote	vase
train	dog	handbag	skateboard	banana	chair	keyboard	scissors
truck	horse	tie	surfboard	apple	couch	cell phone	teddy bear
boat	sheep	suitcase	tennis racket	sandwich	potted plant	microwave	hair drier
traffic light	cow	frisbee	bottle	orange	bed	oven	toothbrush

Figure 7:Classes of COCO dataset

2.3 Model Selection

2.3.1 HARIS Driver Distraction Detection Models

- YOLO v5.

We chose to use YOLOv5 for our project on both distracted driving detection and seatbelt detection. YOLOv5 is a state-of-the-art object detection model known for its high accuracy and real-time performance. It utilizes advanced techniques like anchor boxes and feature pyramid networks. With YOLOv5, we can accurately detect distracted driving behaviors and seatbelts in images or video streams.

The structure of the model allows it to effectively handle complex scenarios, making it suitable for detecting multiple distracted driving behaviors simultaneously.

Additionally, one of the key factors in our decision was YOLO's capability to generate a TensorFlow Lite (TFLite) file, which is highly advantageous for our project. The TFLite format enables efficient deployment of the model on resource-constrained devices, allowing our app to run smoothly and effectively on a wide range of platforms.

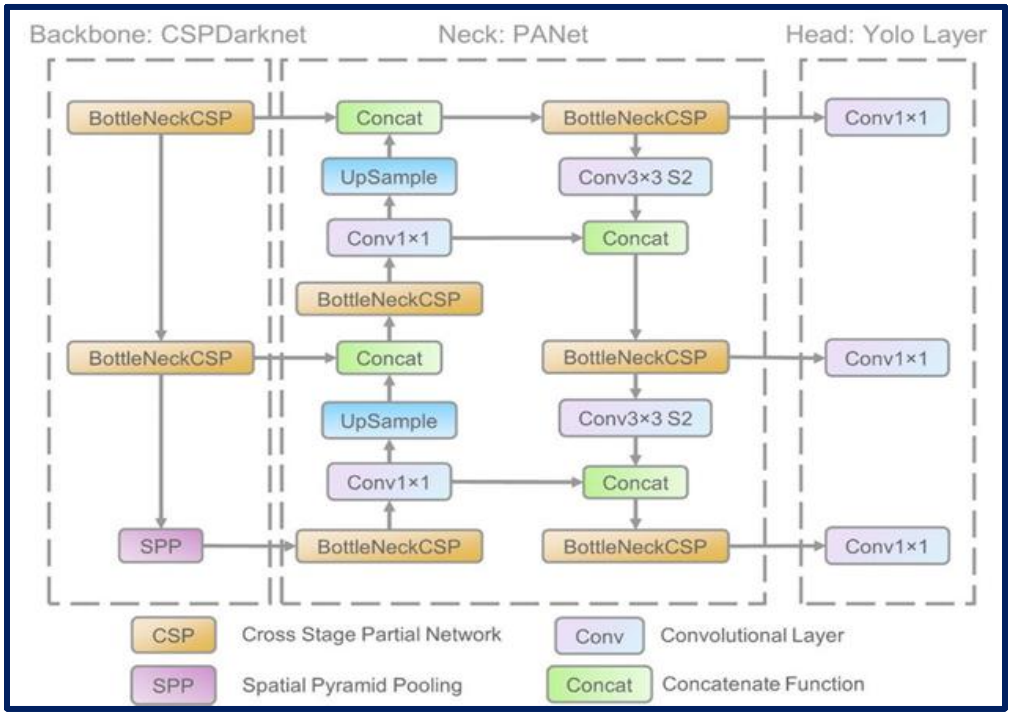


Figure 8:YOLOV5 model's internal structure

The YOLOv5 model shows in Figure 8 used for detecting driver behavior follows a specific structure. It takes input images or frames of a video as its input. These images capture the driver's actions and behaviors inside the vehicle. The model then processes these images through a series of convolutional layers, which extract relevant features and spatial information from the input data.

The output of the YOLOv5 model is a set of detected behaviors. In fact, the model was trained to recognize and classify 12 different kinds of behaviors based on the dataset. These behaviors can include actions such as safe driving, texting, talking on the phone, drinking, operating the radio, yawning, and more. The model assigns a probability score to each detected behavior, indicating the confidence level of the prediction.

- Neural network model

As it is shown in Figure 9, detecting whether a seatbelt is worn is a complex task due to the potential for various actions and occlusions that can lead to inaccurate classification. Factors such as drivers holding objects close to their bodies or wearing clothing resembling seatbelts can create challenges in distinguishing between genuine seatbelt usage and false positives. Additionally, lighting conditions, camera angles, and image quality variations in the car environment further complicate the task. Achieving an effective and accurate seatbelt detector requires robust algorithms that can reliably differentiate true seatbelt usage from misleading visual cues.

Figure 9:Seatbelt misclassified as worn by YOLO due to a Similar Visual Features

Thus, an algorithm that detects the presence of seatbelt is not enough, it is necessary to meticulously determine if it is worn or not. Consequently, in addition to YOLOv5 for seatbelt detection, we integrated a separate neural network model for seatbelt wearing classification. The classification model allows for fine-grained analysis, specifically distinguishing between fastened and unfastened seatbelts.

Table 3: Keras NN model for seatbelt wearing classification task.

Layer	Output Shape	Parameter
Sequential_1	(None,1280)	410208
Sequential_3	(None,2)	128300
Total params	538,508	
Trainable params	524,428	
Non-trainable params	14,080	

In fact, we employed two models to accurately identify instances of seatbelt wearing. Firstly, we utilized YOLOv5, a powerful object detection model, to detect the relevant region within the image that includes the driver and the seatbelt. Once this region was identified, we cropped the image accordingly based on the output from YOLOv5. Next, the cropped images were fed into a separate Keras neural network model specifically designed for seatbelt classification. This model analyzed the cropped images to determine whether a seatbelt was worn or not.

The neural network depicted in Table 3 consists of two layers: an input layer and an output layer. The input layer contains 1280 neurons, which receive the input data or features. The output layer, on the other hand, is composed of 2 neurons, corresponding to the number of classes in the seatbelt wearing classification task. Each neuron in the output layer represents a specific class and provides the probability or score associated with that class.

2.3.2 HARIS Driving Behavior Detection and car plate recognition Models.

- YOLO v8(car behavior model)

Released at the beginning of 2023, YOLOv8 convolutional neural network architecture is the latest generation of the You Only Look Once (YOLO) networks, in particular, and is currently the most efficient model in tasks such as classification, detection, and segmentation of objects [11], In object detection tasks, YOLOv8 can achieve superior results and faster processing times than other models thanks to its combination of optimization techniques and improvements. YOLOv8 utilizes advanced techniques in the object detection field such as decoupled head and anchor-free detection. Our model architecture consists of Backbone, Neck, and Head, as shown in Figure 10.

We train different model sizes from yololv8 on Road Lane Instance Segmentation Image Dataset such as yolov8- m - medium, l - large, and x - extra-large to accurately detect and segment road lanes in real-time video streams or images.

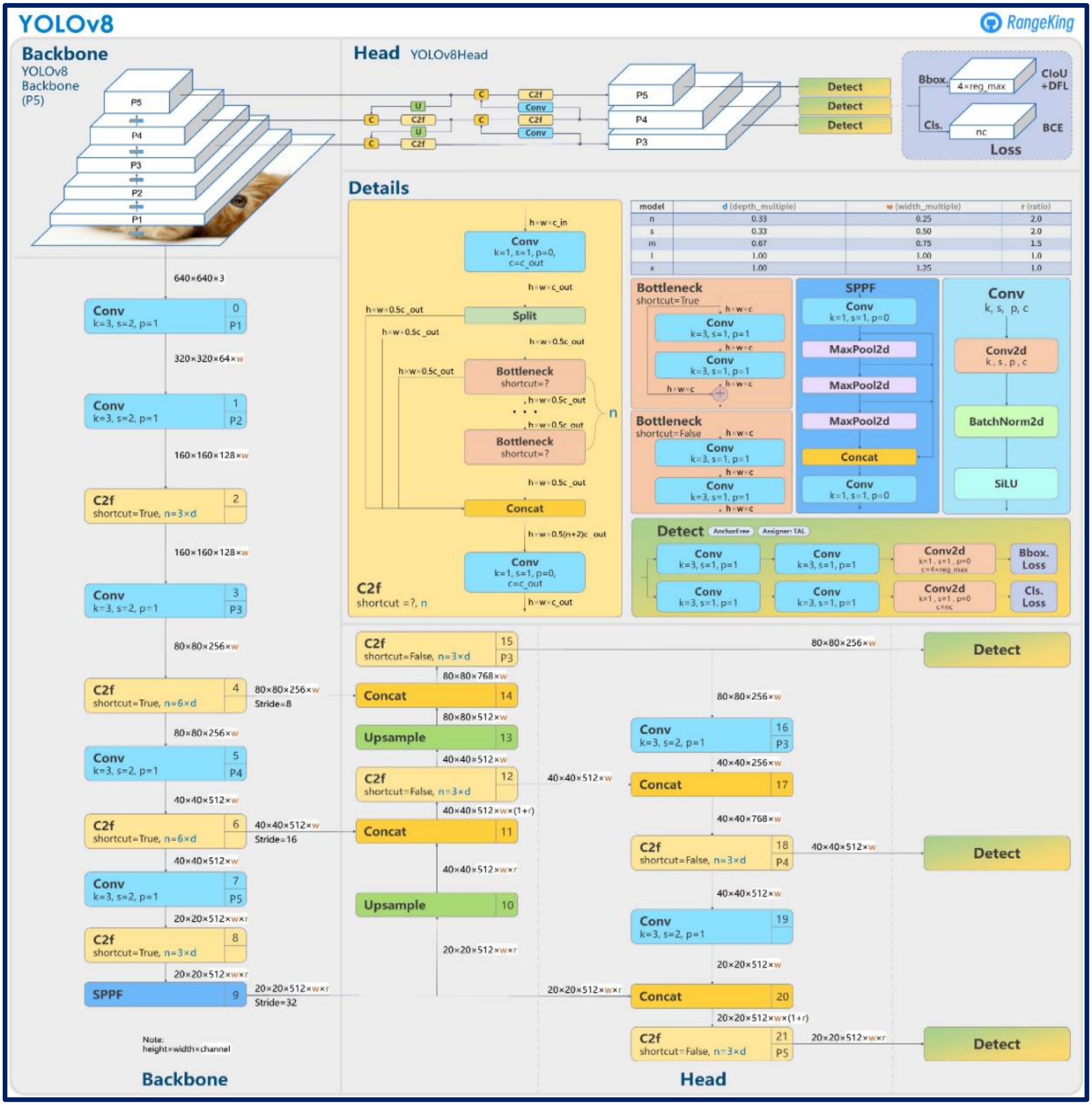


Figure 10: The architecture of YOLOv8 algorithm model. The overall architecture is divided into four parts, including Backbone, Neck, Head, and Loss.

- Byte Tracking

Multiple object tracking (MOT) is a computer vision task that involves tracking the movements of multiple objects over time in a video sequence. The goal is to determine the identity, location, and trajectory of each object in the video, even in cases where objects are partially or fully occluded by other objects in the scene.

The tracking task is divided into two parts. The first one is detecting the object and the second one is using an algorithm to track that object in subsequent frames. For Car detection we use YOLOv8 [7] and we use ByteTracking to track cars and assign unique ID to each car. By assigning IDs to track cars, it becomes possible to create a comprehensive database of driver behavior. The collected data can be analyzed to identify patterns, trends, and correlations between driving behavior and performance outcomes. This analysis can provide insights into factors that contribute to accidents, near misses, or suboptimal performance, enabling targeted interventions. One of the key advantages of ByteTrack is its ability to track objects in real-time. It is designed to run in real-time on a standard computer or even on mobile devices, and it has been shown to achieve high frame rates and low latency in a range of tracking scenarios.

- Convolutional Recurrent Neural Network (CRNN)

The CRNN is the combination of two of the most prominent neural networks, involves CNN (convolutional neural network) followed by the RNN (Recurrent neural networks) Since the features that need to be extracted from license plate images is extremely limited, we propose a simple CNN architecture for feature extraction in our CRNN model. For the recurrent layers, long short-term memory (LSTM) is used instead of normal RNN to prevent exploding gradient problem. Instead of using one layer of LSTM, we stacked two bidirectional LSTM together to capture more details and to increase the accuracy of per-frame. Figure 11.

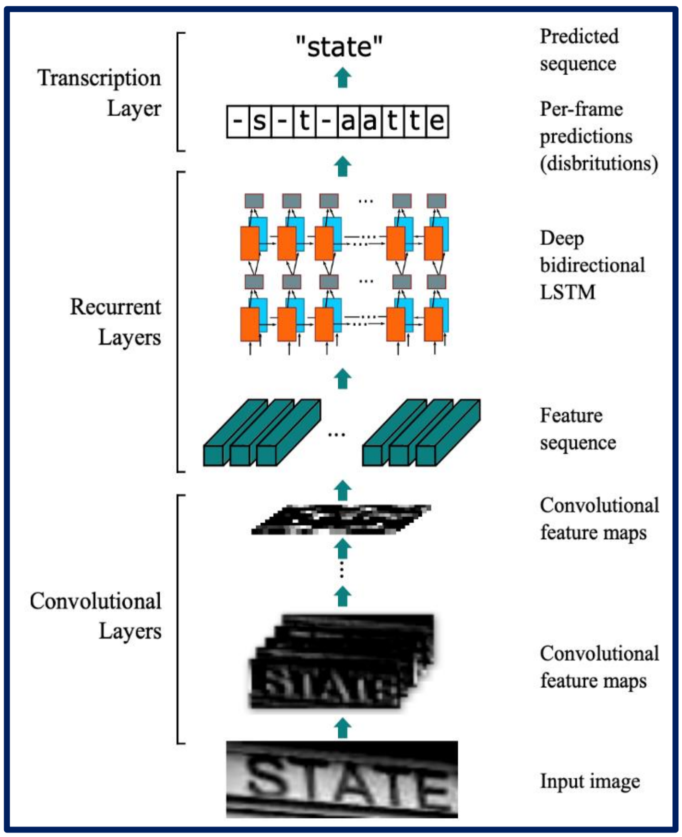


Figure 11:Architecture of CRNN for car plate recognition

Input shape for our architecture having an input image of height 32 and width 128. Here we used seven convolution layers of which 6 are having kernel size (3,3) and the last one is of size (2.2). And the number of filters is increased from 64 to 512 layer by layer. Two max-pooling layers are added with size (2,2) and then two max-pooling layers of size (2,1) are added to extract features with a larger width to predict long texts. Also, we used batch normalization layers after the fifth and sixth convolution layer.

For this CRNN model, we utilized a special loss function called Connectionist Temporal Classification (CTC) loss. Under this loss function, the output of a network is treated as conditional probability distribution over all possible label sequences. An objective function can be derived from this conditional probability to directly maximize the probabilities of the correct labelling. Backpropagation through time is possible because the objective function is differentiable. This loss function solves the challenge of single characters spanning multiple time steps, which would otherwise require additional processing steps. By incorporating a blank symbol and summing over possible alignments.

2.3.3 Proposed model:

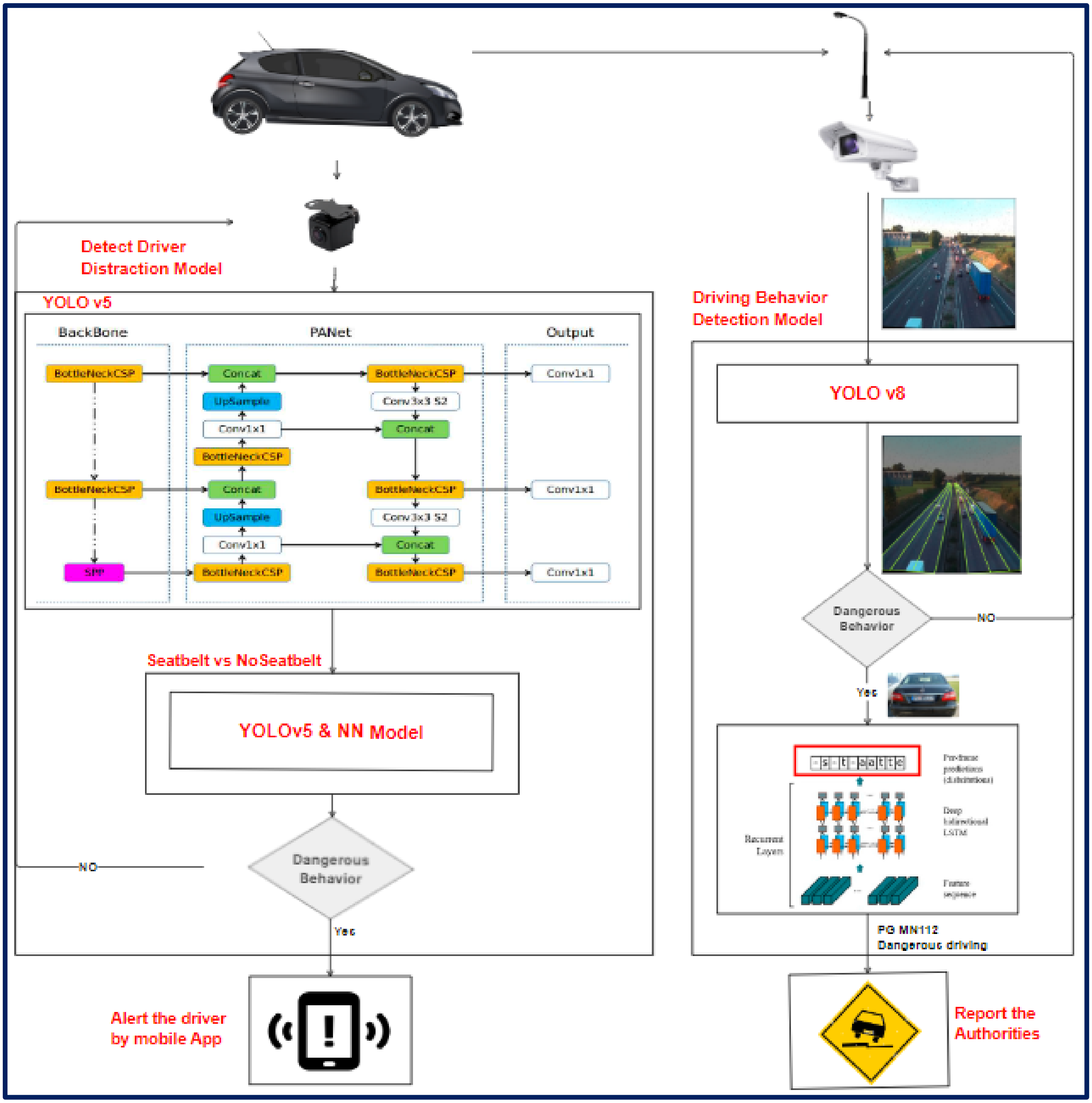


Figure 12: HARIS full system flowchart

HARIS is a comprehensive system comprising two subsystems designed to enhance driver safety and monitor driving behavior. Figure 12

The first subsystem is the "HARIS Driver Distraction Detection Model," which utilizes the YOLOv5 model to detect and classify multiple distraction classes. The proposed flowchart for this subsystem involves input video frames, which are processed by the YOLOv5 model. The model identifies and categorizes distractions such as texting, talking on the phone, or drinking. and then the input video frames processed using the YOLOv5 model, which has been trained to detect and locate seat belts within the frames. Once the seat belts are detected, a neural network classifies the seat belt status. The main action involves sending an immediate alert to the driver through a mobile application.

The second subsystem is the "HARIS Driving Behavior Detection Model." This subsystem utilizes the YOLOv8 model combined with the ByteTrack algorithm to detect car behavior on the road. The flowchart starts with input video frames from road camera, which are analyzed by the YOLOv8 and ByteTrack models. The models detect various driving behaviors, such as aggressive lane changes. If the detected behavior is deemed dangerous, the flowchart proceeds to the next step.

Next, the system employs a CRNN (Convolutional Recurrent Neural Network) model and OCR (Optical Character Recognition) techniques to detect the license plate of the dangerous vehicle. The CRNN model is trained to recognize and extract characters from the license plate, while the OCR algorithm processes the extracted characters for further analysis. Based on the identified dangerous behavior and the license plate information, the system takes appropriate action. This may include notifying the government or relevant authorities about the dangerous behavior, by generating reports to facilitate timely intervention and ensure road safety.

2.4 Model Training and Evaluation

2.4.1 HARIS Driver Distraction Detection Training and Evaluation

The driver distraction detection tasks involved two separate models: one for driver behavior detection and another for seatbelt wearing classification. As a result, the training phase for each model was carried out independently using distinct datasets.

- Driver behavior detection using YOLOv5:

For this part of our project, we utilized a dataset from Roboflow, a platform for managing and annotating datasets. Instead of manually downloading the dataset, we integrated it into our project on Google Collab using the Roboflow API. Next, we cloned the YOLOv5 Ultralytics repository to access the pre-existing YOLOv5 model architecture.

Then, we customized the YOLOv5 model architecture for our project. We made adjustments to the number of classes, anchor boxes, and other architectural parameters to suit our specific requirements. These modifications were implemented to enhance the performance and accuracy of the YOLOv5 model in detecting driver behaviors.

We trained our YOLOv5 model using the dataset and performed a total of 100 epochs to optimize its performance. During the 99th epoch of training, the YOLOv5 model achieved the following metrics: it had a box loss of 0.02382, an object loss of 0.01707, and a class loss of 0.01311.

For the evaluation metrics, when considering all classes, the model achieved a precision (P) of 0.945, recall (R) of 0.944, mean average precision (mAP) at 0.5 intersection over union (IoU) of 0.973, and mAP at the range of IoU from 0.5 to 0.95 of 0.751. These metrics were computed based on 1000 test images containing a total of 1711 instances of objects.

- Seatbelt wearing classification using YOLOv5 and NN model:

The seatbelt wearing detection task involved utilizing a pre-trained YOLOv5 model obtained from Sanketh Jain's GitHub repository [13]. This model was trained using the dataset previously described in the report. It was responsible for detecting seatbelt regions in the images. Additionally, a separate pre-trained neural network model was employed for the classification task, determining whether the detected seatbelt was worn or not.

After conducting extensive testing on numerous custom images and videos captured from various camera angles, it was observed that the performance of the seatbelt detection model was satisfactory. As a result, no further training was performed on the model. Its accuracy and reliability in identifying seatbelt regions and classifying their status as either worn or not worn were consistently maintained, reinforcing the decision to refrain from additional training.

After the training phase, both the driver behavior detection model and the seatbelt wearing classification model were deployed using edge AI techniques. This involved converting the trained models into the TensorFlow Lite (tfLite) format, which is optimized for running on edge devices with limited resources. The tfLite files generated from each model can be easily integrated into a device or mobile app, enabling real-time inference and on-device processing.

2.4.2 HARIS Driving Behavior Detection and car plate recognition Training and Evaluation.

For the car plate detection, we use our custom YOLOv5-m model to train on Car License Plate Detection Dataset. In our custom model we reduce the number of blocks in each stage compared to YOLOv5 to further reduce the computational cost. Additionally, we adopt the Spatial Pyramid Pooling - Fast (SPPF) module in Stage 4, which is an improvement from Spatial Pyramid Pooling (SPP) [12] to improve the inference speed of the model. These modifications lead to our model with a better learning ability and shorter inference time. The architectural model must be tuned in order to achieve the requisite mean average precision. Batch size is 32 and epochs is 10 for the training.

2.5 Experimental Setup:

As a team, we successfully implemented HARIS in Python using Keras with TensorFlow backend. We conducted the training and evaluation on an NVIDIA P100 GPU with 12 GB of RAM in the Google Collaboratory environment. Additionally, for the Detect Driver Behavior by Image model, we utilized the PyTorch framework, leveraging the power of the same GPU configuration. To achieve accurate and efficient object detection, we employed advanced models including YOLOv5 and YOLOv8. Furthermore, we utilized Roboflow for data annotation and preprocessing to enhance the training process. This combined approach allowed us to achieve robust and reliable detection of driver distractions and various driving behaviors, contributing to improved road safety and a deeper understanding of driver actions.

3.Results and Analysis

3.1 HARIS Driver Distraction Detection Results

The results of training and testing the YOLOv5 model for distracted driver behavior detection and the YOLOv5 combined with a neural network (NN) in Keras for seatbelt wearing classification were highly promising. Both models achieved impressive accuracy levels, accurately identifying the respective behaviors. The models also demonstrated satisfactory performance when tested with varying camera angles and image quality inputs. Although there were minor limitations in certain scenarios, overall, the results showcased the models' effectiveness in detecting and classifying driver behaviors, contributing to improved safety and awareness on the road.

- Comparison with Baseline or Prior Work

In the context of dataset classes, the first-term model was trained on a dataset that included fewer driver behaviors compared to the second term. Additionally, the first term model included multiple classes that were not commonly observed behaviors among all drivers, such as "texting left" and "texting right," which could be considered redundant. In the second term model, these two classes were combined into a single class, which improved the specificity and accuracy of the predictions Table 4. This refinement is crucial for targeting the correct results and ensuring that the model focuses on the most relevant and practical driver behaviors.

Table 4: Comparison of Detected Behaviors between Previous and Current Models

Behaviors detected by the previous model	**Behaviors detected by the current model**
C0 - Safe driving.	c0 - Safe Driving.
C1 -Texting – right.	c1 – Texting.
C2 -Talking on the phone – right.	c2 - Talking on the phone.
C3 -Texting – left.	c3 - Operating the Radio.
C4 -Talking on the phone – left.	c4 – Drinking.
C5 -Operating the radio.	c5 - Reaching Behind.
C6 -Drinking.	c6 - Hair and Makeup.
C7 -Reaching behind.	c7 -Talking to Passenger.
C8 -Hair and makeup.	d0 -Eyes Closed.
C9 - Talking to passenger.	d1- Yawning.
	d2 - Nodding Off.
	d3 - Eyes Open.

In the context of model performance, the previous-term model consists of a custom CNN model which achieved a high accuracy of 99% in the testing phase. However, this model faced challenges when dealing with custom images taken from various camera angles and with different resolutions. In the current project, we adopted the YOLO model, which effectively addressed the limitations of the previous approach. Although the YOLO model did not surpass the test accuracy of the previous model, its performance on real-world images made it the preferred choice. The YOLO model demonstrated superior adaptability and robustness, making it well-suited for our project's requirements.

3.1.1Evaluation Metrics

Mean average precision (mAP) is a preferred metric over other accuracy metrics in the context of YOLOv5 and object detection models. It considers both the precision and recall for each object class, handles multiple object classes effectively, incorporates confidence scores, and evaluates performance at different IoU thresholds. mAP provides a comprehensive evaluation of localization and classification accuracy, making it a more informative and nuanced metric for assessing the performance of object detection models the performance curve of the result shows in Figure13:

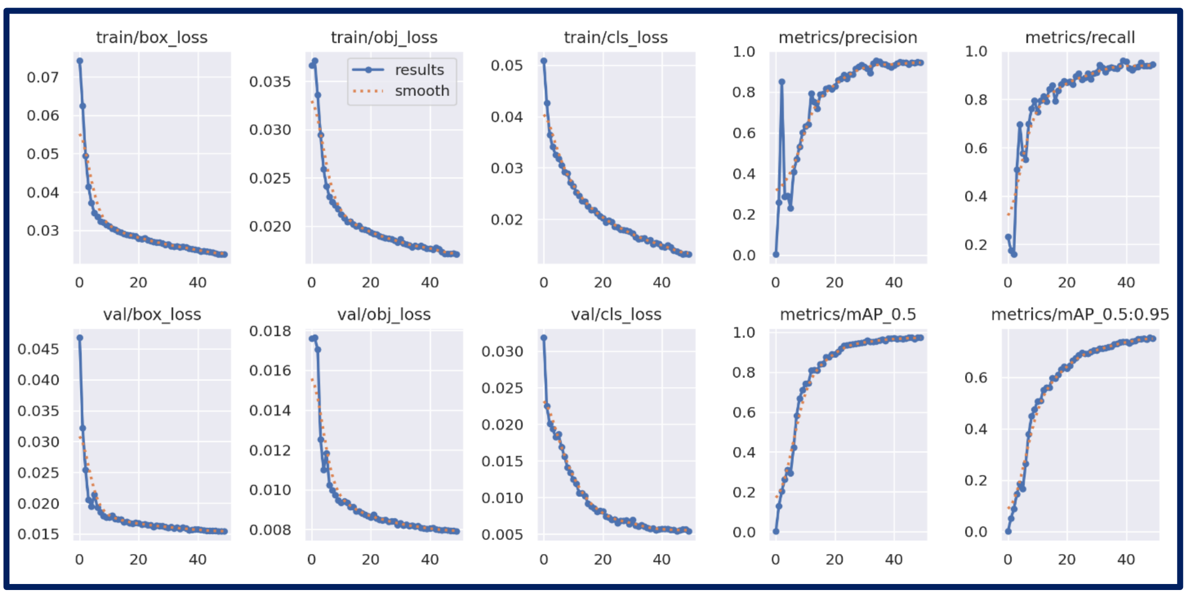


Figure 13:Training and validation performance of YOLOv5m model for driver behavior detection

Table 5:Driver behavior detection model evaluation results

Precision (P)	Recall (R)	mAP	IoU
0.945	0.944	0.5	0.973

Table 5 above provides a summary of the metrics results obtained from yolov5 model. These metrics are commonly used to evaluate the performance of models, algorithms, classification tasks, or predictive modeling.

- Precision: Precision is the metric that measures the accuracy of positive predictions made by a model. In the context of YOLOv5, precision represents the proportion of correctly predicted bounding boxes (detections) out of all the predicted bounding boxes. It indicates how well the model avoids false positives.

- Recall: Recall, also known as sensitivity or true positive rate, measures the ability of a model to correctly identify all positive instances. In YOLOv5, recall represents the proportion of correctly predicted bounding boxes out of all the ground truth bounding boxes. It shows how well the model avoids false negatives.

- Intersection over Union (IoU): IoU is the metric that evaluates the overlap between the predicted bounding boxes and the ground truth bounding boxes. It calculates the ratio of the intersection area to the union area between two bounding boxes. In YOLOv5, IoU is used to determine whether a prediction is a true positive or a false positive based on a specified threshold.

- Mean Average Precision (MAP): mAP is a widely used metric to evaluate object detection models like YOLOv5. It combines precision and recall across different IoU thresholds to assess the overall performance of the model. It calculates the average precision at different recall levels and then takes the mean across all classes. Higher mAP values indicate better detection accuracy. Precision: Precision is a metric that measures the accuracy of positive predictions made by a model. In the context of YOLOv5, precision represents the proportion of correctly predicted bounding boxes (detections) out of all the predicted bounding boxes. It indicates how well the model avoids false positives.

3.1.2 Visual Representations

Figure 14:Instances of results obtained from testing the first mode.

In the obtained Figure 14 processed by the YOLOv5 model, it shows that the model effectively detected driver behaviors in the images, including safe driving, talking to passengers, and drivers with their eyes open and eyes closed also. These results demonstrate the model's ability to detect various frequent distracted behaviors noticed among drivers.

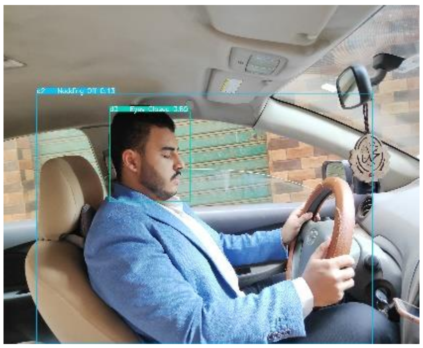
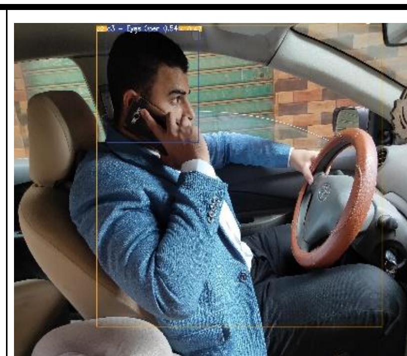
		
Nodding off Eyes closed	Talking to passenger Eyes open	Talking on the phone Eyes open

Figure 15: Instances of results obtained from testing on real data.

These results demonstrate the capability of YOLOv5 to accurately detect and classify various driver behaviors in real-world scenarios. The figure showcases instances of results obtained from testing real data using YOLOv5 for the task of detecting driver behavior. The system is trained to identify specific behaviors such as "Nodding off," "Eyes closed," "Talking to passenger," "Eyes open," and "Talking on the phone."

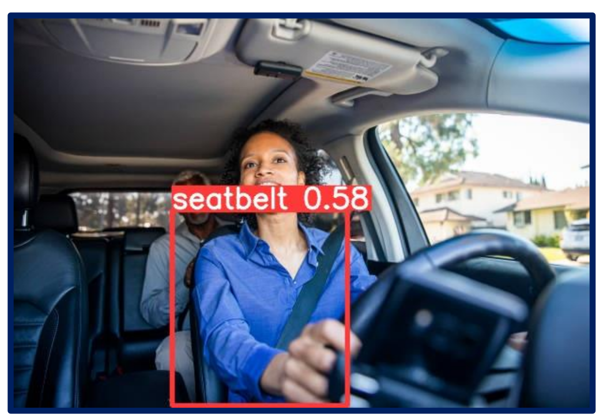
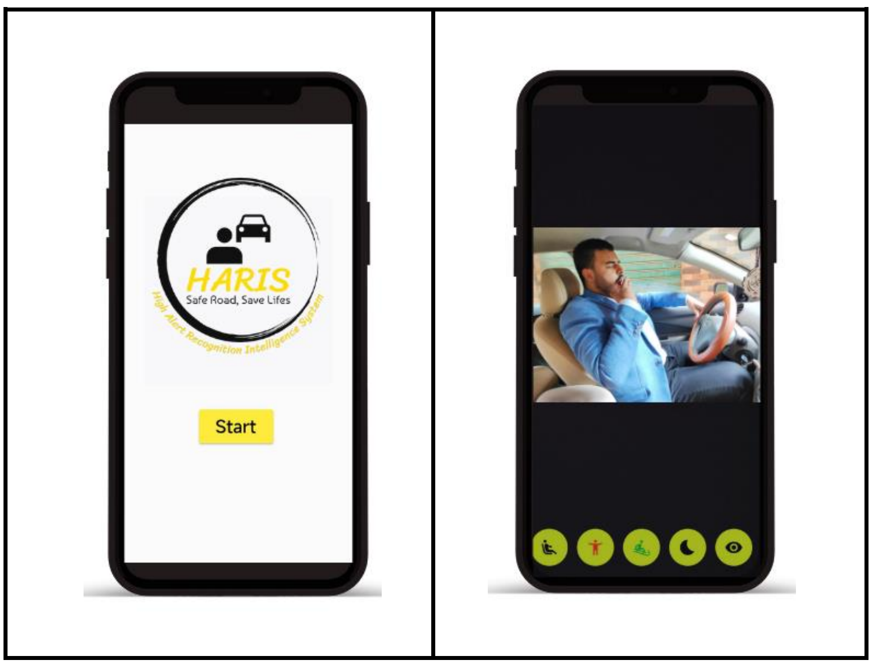


Figure 16:Instances of results obtained from testing the second model.

This Figure 16 shows the capability of the Seat belt detection model to accurately detect and classify the cases if the driver wears a seatbelt or not.

3.1.3 Simple Mobile Application

We developed a simple mobile app for the HARIS Distracted Driver Behavior serves as in promoting driver safety and mitigating dangerous behavior. With its intuitive interface and real-time functionality, the app provides drivers with timely alerts and notifications to prevent potential accidents. By integrating with the HARIS system, the app leverages advanced computer vision techniques to analyze driver behavior and driving patterns, identifying instances of distraction or unsafe actions. Through the app, drivers receive immediate warnings and guidance, helping them make informed decisions and stay focused on the road. This mobile app acts as a dependable companion, enhancing driver awareness and creating a safer driving experience for all.



The second page of the mobile app for the HARIS project features a powerful camera functionality that captures live video of the driver's behavior. This innovative feature enables real-time monitoring and analysis of the driver's state of distraction. By utilizing computer vision algorithms, the app quickly processes the captured images and provides an instant assessment of the driver's level of distraction. Whether it is detecting texting, talking to passengers, or other forms of distractions, the app delivers accurate and real-time feedback to the driver. This camera page serves as a valuable tool in promoting self-awareness and encouraging responsible driving habits, contributing to a safer driving environment.

3.2 HARIS Driving Behavior Detection and car plate recognition Results.

The results of employing yolov8 for car and road lane detection and ByteTracking for car tracking and behavior analysis, demonstrated highly accurate results. The model effectively detected and tracked cars on the road, enabling comprehensive analysis of driving behavior. The integration of yolov8 and ByteTracking yielded precise results with an impressive level of accuracy.

Moreover, the car plate recognition component of HARIS, utilizing the CRNN model for OCR, achieved high accuracy in recognizing and extracting information from license plates. The CRNN model exhibited superior performance, delivering accurate results with a high degree of line accuracy.

While the overall results of the HARIS system were highly accurate and reliable, there are a few limitations to consider. Firstly, the performance of the models may be influenced by environmental factors such as occlusions, and variations in camera perspectives. Additionally, the system's accuracy may vary in challenging scenarios with complex traffic situations or unconventional driving behaviors. Continued refinement and fine-tuning of the models can help address these limitations and further enhance the system's performance.

3.2.1 Evaluation Metrics

YOLOv8 for segmentation commonly used evaluation matrices include Intersection over Union (IoU), Average Precision (AP), and mean Average Precision (mAP) with a specific threshold 0.5 and 0.5-0.95.

IoU measures the overlap between predicted and ground truth bounding boxes, AP calculates the precision-recall curve for each class, and mAP with a specific threshold, such as IoU threshold of 0.5 and IoU threshold of 0.5-0.95. This metric measures the average precision across different object categories as shown in Figure 17, considering the detection performance above the specified threshold. mAP with a threshold provides a comprehensive evaluation of the object detection performance of YOLOv8.

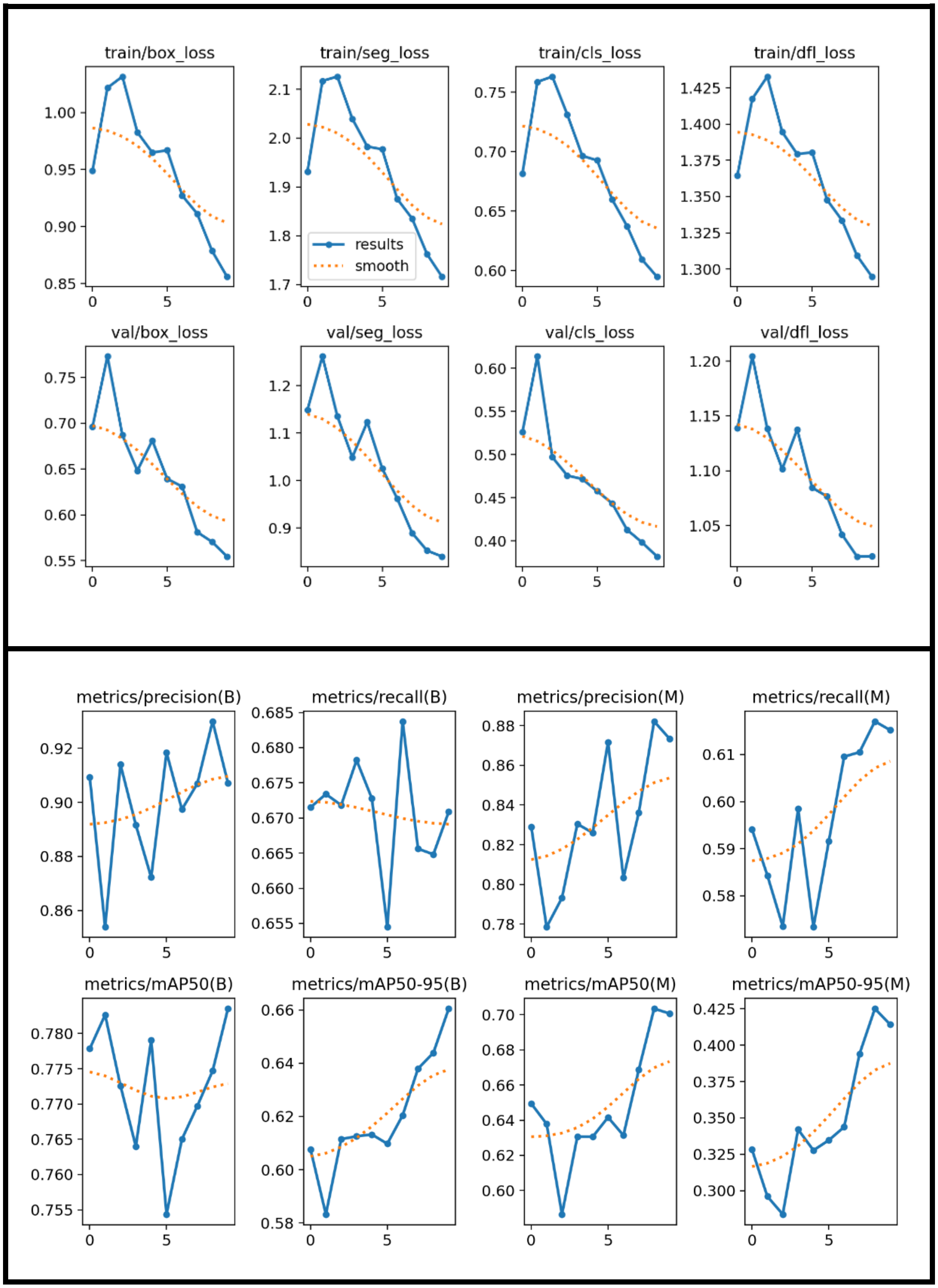


Figure 17:Training and validation performance of YOLOv8-x model for road lane segmentation

For car plate detection One of the most important measures in evaluation car plate detection mAP. For car plate detection One of the most important measures in evaluation car plate detection MAP. The area under the precision-recall curve is used to compute the average precision, where Precision=TP/(TP+FP) and Recall=TP/(TP+FN). Intersection over Union (IoU) is a classical metric for evaluating the performance of the model for object detection. It calculates the ratio of the overlap and union between the generated candidate bounding box and the ground truth bounding box figure 18.

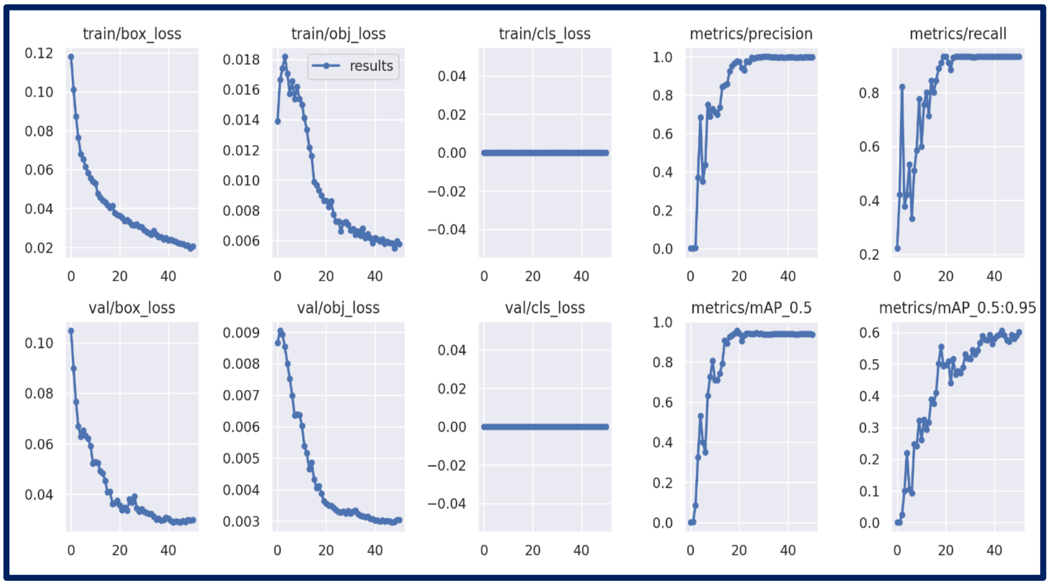


Figure 18:Training and validation performance of YOLOv5m model for car plate detection.

In car plate Number and character Recognition we calculated the accuracy of prediction using Line Accuracy 0.91%. A license plate can only be considered correctly labelled if all individual characters in the license plate are predicted correctly. Although we evaluate performing in character level recognition. So, we calculate the character level recognition accuracy figure 19 using the following equation:

$$\text{Character level accuracy} = \frac{\#\ of\ corectly\ label\ character}{\sum \#\ of\ character\ in\ all\ licence\ plates} \times 100\%$$

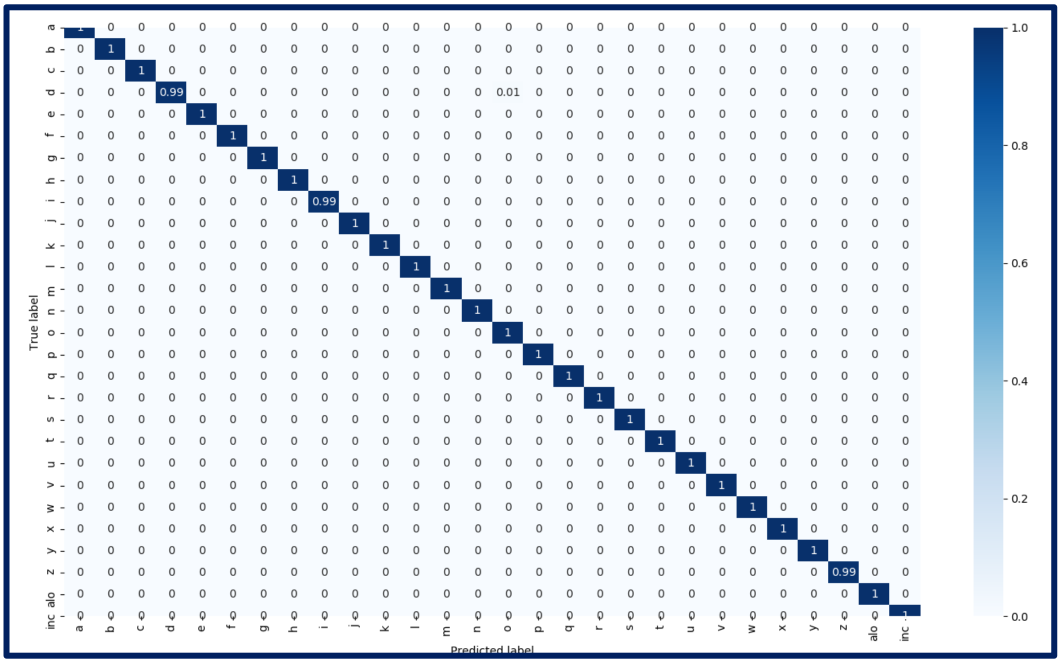


Figure 19:confusion matrix of character lever accuracy

3.2.2 Visual Representations

We present the comprehensive outcomes achieved through the utilization of YOLOv8 for car and road lane detection, tracking, and ID assignment In Figure 20. This advanced algorithm effectively detects and tracks cars on the road, providing valuable information about their positions and movements. By assigning unique IDs to each detected car, it becomes possible to precisely track and analyze individual vehicles over time. Such detailed tracking and identification capabilities enhance various applications, including traffic management.

Figure 20:the results of car and road lane detection and tracking

Figure 21 illustrates a specific scenario where a car engages in reckless driving by changing lanes, effectively represented by a clearly distinguishable red bounding box. This visualization plays a vital role in promoting road safety by raising awareness, facilitating analysis, and informing the development of effective measures to address reckless driving practices.

Figure 21:illustrates a specific scenario where a car in reckless driving by changing lanes.

In Figure 22, the results of road line detection using the YOLOv8 and ByteTrack models are showcased, demonstrating the accurate detection of each road within the frames. The advanced algorithms employed in these models enable precise identification and tracking of road lines, contributing to improved lane detection and overall road understanding,

Figure 22:demonstrates the output of road lane segmentation achieved using YOLOv8.

The Figure 23 showcases a comprehensive car plate detection system powered by YOLOv5. After successfully detecting license plates, Optical Character Recognition (OCR) is applied to recognize the characters on the plate. The recognized characters are returned in text format, maintaining their original order.

Figure 23:result of car plate recognition

Figure 24 illustrates the successful implementation of car plate detection using the YOLOv5 algorithm. The system effectively identifies and localizes license plates on vehicles, as demonstrated by the precise bounding box annotations

Figure 24:Visual Representations of car plate detection

In Figure 25, the car plate numbers are checked and compared to the wanted database of the authorities using the YOLOv5 model for car plate detection (yellow boxes for the safe plates and green for the wanted plates). Subsequently, the plate numbers are recognized and matched against the database using the CRNN model for plate number recognition. This integrated approach ensures accurate verification and enhances security measures.

Figure 25: result of detecting wanted car plates.

4. Conclusions and Recommendations

In conclusion, the High Alert Recognition Intelligent System (HARIS) represents a groundbreaking AI-based solution that utilizes computer vision techniques to analyze and interpret driver behavior and driving behavior in real-time. By integrating in-car and road cameras, HARIS effectively detects and classifies various driver distractions and unsafe driving behaviors, contributing to improved road safety. The implementation of the HARIS Driver Distraction Detection model, powered by YOLOv5, enables accurate identification and classification of driver distractions, such as texting or talking to passengers. This capability allows for timely notifications and alerts to be sent to drivers, reducing the risks associated with distracted driving.

Additionally, the HARIS Driving Behavior Detection model, utilizing YOLOv8 and ByteTrack algorithms, effectively monitors car behaviors and road lanes. This model identifies unsafe driving practices, including aggressive lane changes, promoting a safer driving environment. Furthermore, the integration of YOLOv5 and a Convolutional Recurrent Neural Network (CRNN)-based OCR model empowers HARIS to accurately detect and recognize car license plates. This capability aids in identifying potentially dangerous vehicles, adding an extra layer of safety on the road. Overall, the development and implementation of HARIS presents a significant advancement in leveraging AI-powered technology to enhance road safety. The system's real-time analysis and detection capabilities have the potential to significantly reduce accidents caused by driver distractions and unsafe driving behaviors. While HARIS shows promising results, it is important to acknowledge its limitations, such as variations in lighting conditions and the quality of captured data. Continued research and advancements in this field will refine HARIS's capabilities, contributing to the development of more intelligent and safer transportation systems.

In summary, HARIS represents a transformative solution that aims to revolutionize road safety. By accurately detecting driver distractions, monitoring driving behavior, and identifying potential risks, HARIS contributes to creating a safer driving environment for all road users. With ongoing advancements, HARIS holds immense potential for improving road safety standards and promoting responsible driving practices in the future. To enhance HARIS and improve road safety, we need to:

1. Allocate more resources for training and data collection to improve the performance of the Driver Distraction Detection model.

2. Collect a comprehensive dataset for the Seat belt Detection model and explore alternative algorithms to enhance its results.

3. Combine the Driver Distraction Detection and Seat belt Detection models to provide a more comprehensive analysis of distractions and seat belt usage.

4. Acquire a real dataset and upgrade road cameras to improve the accuracy of the Driving Behavior Detection model in detecting driving behaviors.

5. Identify and incorporate additional reckless driving behaviors by collaborating with experts and conducting continuous research. These efforts will enhance HARIS's ability to promote responsible driving practices and contribute to improved road safety.

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